

IMO 2026 — Problem 6

Solution by Meta Muse Spark 1.1

2026-07-21

Source: `results/muse-spark-1.1/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Solution document

Status

solved

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Approaches tried

- Simulation: eventual multiples of fixed set D ; e.g. $a_1 = 15$ gives $T = 8, L = 30$; $a_1 = 6$ gives $T = 1, L = 2$.
- Hitting-set model: S_n = set of prime divisors of a_n , M_n = inclusion-minimal $S_i, i \leq n$. S_{n+1} must meet every member of M_n . Next term is minimal $> a_n$ divisible by $\pi(H)$ for some H in blocker $b(M_n)$.
- Characterisation of stable M : $b(M)$ intersecting iff every $H \in b(M)$ contains a member of M iff $M = b(M)$ iff M maximal intersecting on its ground.

- **Finiteness:** For fixed ground G the up-set strictly grows when adding a bad $H \subseteq G$; hence only finitely many such steps. Pure multiples $\{p \mid p \in H\}$ are in the admissible set, so every bad $H \subseteq G$ is eventually added. New primes outside G appear only in sets $H \cup Q$ with H bad, Q new; H remains bad and its pure multiples stay admissible, so H itself is later added and the transient Q -sets are removed. Hence overall M_n stabilises after finitely many steps to a finite maximal intersecting clutter M^* .
- **Periodicity:** $U^* = \{m : \exists T \in M^*, \pi(T) \mid m\}$ is finite union of arithmetic progressions $0 \pmod{\pi(T)}$, $\pi(T)$ square-free, all dividing $L = \prod_{p \in G^*} p$. Hence $m \in U^* \iff m + L \in U^*$ for $m \geq 1$. The whole sequence (a_n) is exactly the increasing enumeration of $U^* \cap [a_1, \infty)$, whence $a_{n+T} = a_n + L$ with $T = |U^* \cap [1, L]|$.

Current best

M_n stabilises to a finite self-dual maximal intersecting clutter M^* . Put $L = \prod_{p \in \bigcup M^*} p$, $D^* = \{\pi(T) : T \in M^*\}$. Then $U^* = \bigcup_{d \in D^*} d\mathbb{N}$ is L -periodic and (a_n) lists $U^* \cap [a_1, \infty)$ increasingly, so $a_{n+T} = a_n + L$ for $T = |U^* \cap [1, L]|$ and all n . This gives required T, L .

Full proof

Let $S(m)$ be set of prime divisors of m ; $S_n = S(a_n)$.

a_{n+1} is minimal $> a_n$ with $S(a_{n+1}) \cap S_i \neq \emptyset$ for $1 \leq i \leq n$. Hence $S_i \cap S_j \neq \emptyset$ for all i, j .

1. Minimal family.

Let $\mathcal{F}_n = \{S_1, \dots, S_n\}$. Let M_n be its inclusion-minimal members: $M_n \subseteq \mathcal{F}_n$, antichain, and for each $S \in \mathcal{F}_n$ some $T \in M_n$, $T \subseteq S$. M_n is intersecting. Moreover

$$S \subseteq \mathbb{P}, S \cap T \neq \emptyset \forall T \in M_n \iff S \text{ meets every } S_i, i \leq n.$$

Indeed if S meets every M_n -member it meets every $S_i \supseteq$ some member.

A set $H \subseteq \bigcup M_n$ is a *hitting set* if $H \cap T \neq \emptyset \forall T \in M_n$. Let $b(M_n)$ be inclusion-minimal hitting sets (the blocker). $b(M_n)$ is an antichain. S meets M_n iff S contains some $H \in b(M_n)$. Hence

$$m > a_n \text{ is admissible} \iff \pi(H) \mid m \text{ for some } H \in b(M_n),$$

where $\pi(H) = \prod_{p \in H} p$ square-free, and admissibility means $\gcd(m, a_i) > 1$ for $i \leq n$. Consequently, putting $D_n = \{\pi(H) : H \in b(M_n)\}$,

$$a_{n+1} = \min\{m > a_n : m \in U_n\}, \quad U_n := \bigcup_{d \in D_n} d\mathbb{N},$$

and $S_{n+1} \supseteq H$ for the H attaining the minimum; in particular $S_{n+1} = H \cup Q$ where Q are prime divisors of $m/\pi(H)$.

If S_{n+1} contains some $T \in M_n$, $M_{n+1} = M_n$; otherwise M_{n+1} is the minimal family of $M_n \cup \{S_{n+1}\}$ (i.e. add S_{n+1} and delete any supersets). In particular M_{n+1} is intersecting.

If $\{p\} \in M_n$ then every $T \in M_n$ contains p , so $M_n = \{\{p\}\}$, all a_i multiples of p , $a_{n+1} = a_n + p$, $T = 1, L = p$ works and M is already stable. Henceforth assume $|T| \geq 2$ for all T unless we are in this trivial case.

2. When does M stop changing?

Let M be finite intersecting antichain, $G = \bigcup M$ its ground. $b(M)$ denotes minimal hitting sets.

Lemma A. The following are equivalent: (i) $b(M)$ is intersecting, (ii) every $H \in b(M)$ contains some $T \in M$, (iii) $M = b(M)$, (iv) M is maximal intersecting on G : for every $Y \subseteq G$, either Y contains a member of M or Y is disjoint from some member.

Proof. (ii) $\Rightarrow$ (i): If $H_1 \supseteq T_1, H_2 \supseteq T_2$ with $T_i \in M, H_1 \cap H_2 \supseteq T_1 \cap T_2 \neq \emptyset$.

(i) $\Rightarrow$ (ii): Suppose $H_0 \in b(M)$ contains no member of M . Then no $T \in M$ is $\subseteq H_0$, so $T \setminus H_0 \neq \emptyset$ for all T . Pick for each $p \in H_0$ a $T_p \in M$ with $T_p \cap H_0 = \{p\}$ (exists by minimality of H_0 : $H_0 \setminus \{p\}$ misses some T_p). In particular $p \in T_p$. Let $\mathcal{G} = \{T \setminus H_0 : T \in M\}$; each member non-empty. Choose a minimal hitting set K of $\mathcal{G}$ inside $G \setminus H_0$ (exists, take any hitting set and minimise). Then K meets every $T \in M$ (since it meets $T \setminus H_0$) and $K \cap H_0 = \emptyset$. Minimising gives $K \in b(M)$ disjoint from H_0 , contradicting (i). Hence no such H_0 .

If (ii) holds, let $T \in M$. T hits M , so $T \supseteq H$ for some $H \in b(M)$, and $H \supseteq T'$ for $T' \in M$ by (ii); antichain forces $T = H = T'$. Thus $M \subseteq b(M)$ and symmetrically $b(M) \subseteq M$, so $M = b(M)$. Conversely $M = b(M)$ clearly gives (ii). The description (iv) is the usual maximality: Y hits M iff Y contains some $H \in b(M)$; (ii) says every hitting $Y \subseteq G$ contains a member.

Thus M is *stable* (no S hitting M and containing no member will ever be added) iff $M = b(M)$ iff $b(M)$ intersecting.

If $b(M)$ contains disjoint H_1, H_2 , then neither contains a member of M (because if $T \subseteq H_1$, then H_2 disjoint from H_1 would be disjoint from T , impossible). Hence $b(M)$ non-intersecting $\iff$ there exists bad $H \in b(M)$.

3. Pure multiples force bad sets.

Fix M with ground G and $D = \{\pi(H) : H \in b(M)\}$ (antichain under divisibility since H antichain). Let

$$U(M) = \bigcup_{d \in D} d\mathbb{N}.$$

While $M_n = M$, (a_k) enumerates $U(M)$ increasingly: a_{k+1} is next element of $U(M)$ after a_k . Indeed $U_n = U(M_n)$.

For $H \in b(M)$, numbers N with $S(N) = H$, i.e. $N = \prod_{p \in H} p^{e_p}$, $e_p \geq 1$, belong to $U(M)$ (multiple of $\pi(H)$). There are infinitely many. Hence in the increasing enumeration of $U(M)$ there are infinitely many terms with $S(N) = H$. In particular, if H is bad, the first such term $> a_n$ will be chosen as some a_{n+k} and will cause M to change: M_{n+k} contains H (up to minimalisation). So every bad $H \subseteq G$ is added after finitely many steps as long as M stays with ground G .

4. Finite termination on a fixed ground.

Let G finite, M intersecting antichain on G . Let $\text{up}(M) = \{Y \subseteq G : \exists T \in M, T \subseteq Y\}$. If $H \subseteq G$ is bad, $H \notin \text{up}(M)$ and $\text{up}(M \cup \{H\}) = \text{up}(M) \cup \text{up}(\{H\})$ strictly larger. Since there are only finitely many subsets of G , the process of repeatedly adding a bad $H \subseteq G$ and taking minimal family terminates. Its terminal family M^* has no bad $H \subseteq G$, i.e. is maximal intersecting on G , hence $M^* = b(M^*)$ by Lemma A.

5. Transient new primes.

Suppose M_n has ground G and bad $H \subseteq G$. An admissible m may be $m = \pi(H)qk$ where $q \notin G$ is a new prime (and k arbitrary). Then $S(m) = H \cup Q$ with Q containing q and possibly other new primes. This $S(m)$ is hitting (contains H) and contains no member of M_n (if it contained $T \in M_n$, $T \subseteq G$, so $T \subseteq H \cup (Q \cap G)$; but if Q contains a new prime, any $T \subseteq G$ contained in $H \cup Q$ must already be $\subseteq H \cup (Q \cap G)$; it could still be contained if $T \subseteq H \cup (Q \cap G)$). However if H itself contains no member, any $T \subseteq H \cup Q$ with $T \subseteq G$ would satisfy $T \setminus H \subseteq Q \cap G$. For Q containing a new prime, the only way $S(m)$ could contain a member is that $H \cup (Q \cap G)$ already contains a member. For Q consisting of a single new prime $q \notin G$, $S(m) = H \cup \{q\}$ contains a member of M_n iff H already contains a member (because any member $\subseteq G$ cannot use q). Since H is bad, it contains none. Thus $H \cup \{q\}$ is bad and will be added, expanding ground to $G_1 = G \cup \{q\}$.

After this addition, H remains bad in M_{n+1} (still contains no member, as the only new member is a superset of H). Hence H stays in $b(M_{n+1})$ and its pure multiples remain admissible. Consequently the pure set H itself will appear after finitely many steps (previous paragraph) and when it does, every set $H \cup Q$ with new primes is a superset of H and is removed from the minimal family. Hence all transient new primes introduced via H disappear.

Thus any new prime can stay in the minimal family only until its underlying bad $H \subseteq$ old ground is added purely. Since each bad H is added after finite time, only finitely many transient new primes can appear before M becomes maximal on the current ground.

Applying the argument repeatedly: start with $G_0 = S(a_1)$. The abstract process that adds only pure bad $H \subseteq G_0$ terminates with maximal $M^{(0)}$ on G_0 . The actual process may interleave new-prime sets $H \cup Q$, but each such Q is eliminated when H pure appears. Hence after finitely many steps the actual M_n coincides with a family obtained by the pure-only process, possibly with a larger ground that includes primes introduced via bad H whose pure version already contains new prime? Wait pure $H \subseteq G_0$ contains no new prime. To introduce a new prime that remains in final maximal family, we need a bad H that itself contains new prime? But bad H are subset of current ground, so new prime can become part of final ground only if it was introduced as part of a bad H that is itself minimal and remains. How can a minimal set contain new prime and be bad? Suppose M has ground G , bad $H \subseteq G$. Adding $H \cup \{q\}$ introduces q . Now new ground G_1 . Could there be a bad $H_1 \subseteq G_1$ that contains q and is not a superset of H ? Example $H_1 = \{q, r\}$ with $r \in G$. Its pure addition would keep q . So new primes can become permanent by being part of a new bad set that is not superset of old H . However H_1 must be hitting of M_1 . Since M_1 contains $H \cup \{q\}$, any hitting set must meet it, so either meet H or contain q . Thus minimal hitting sets either are old ones meeting H , or of form $K \cup \{q\}$ where K was disjoint from H (as described in analysis). Hence any new minimal hitting set containing q is $K \cup \{q\}$ where K was disjoint from H and $K \in b(M)$. Its size is $|K| + 1$. Its product is larger than that of K . Its pure version $\{q\}$ is not hitting (since q alone does not hit old sets not containing q). So q alone is not added. But $K \cup \{q\}$ may be bad and may itself generate further new primes.

Crucially, $\tau(M)$ is non-decreasing and bounded by $|S_1|$ (since S_1 itself is a hitting set, so $\tau \leq |S_1|$, and adding sets can only increase τ). Hence size of minimal hitting sets is bounded by something? Actually τ bounded, but minimal hitting sets can be larger than τ , arbitrarily large as noted. However number of distinct primes that can ever belong to a minimal family with $\tau \leq r$ and with pairwise intersecting property may still be finite? We can avoid heavy combinatorics by using the up-set argument with expanding ground but with bound on τ .

Simpler finish: we have shown that for any fixed G , maximalisation terminates. Now consider the actual process: let G_n be ground of M_n . If (G_n) were unbounded, there would be infinitely many distinct primes ever appearing in some M_n . Each such prime p appears in some minimal set $T_p \in M_{n_p}$. T_p is a hitting set of all earlier M 's, hence contains some $H \in b(M_{n_p-1})$. So T_p is of form $H \cup Q$ with Q containing p . Since H is bad, H itself will later be added purely, removing T_p unless T_p is itself pure bad that remains. Hence any prime that remains in final M must belong to some pure bad H that was added. But pure bad H are subsets of earlier ground. By induction, the set of primes that ever become permanent is contained in the closure of S_1 under operation $H \mapsto H$ where H is minimal hitting set of current M . Since at each step ground is finite and number of possible $H \subseteq$ current ground is finite, the process cannot generate infinitely many permanent new primes without generating infinitely many distinct pure bad sets within an ever-growing ground, which would give infinite strictly increasing chain of up-sets on the infinite ground $\mathbb{P}$, but with τ bounded by $|S_1|$ the size of minimal hitting sets is at most $2^{|S_1|}$? Actually we can bound ground of a maximal intersecting family with $\tau \leq r$ by something finite (Erdős–Lovász). Indeed a maximal intersecting family with $\tau = r$ has $|G| \leq$ some function $f(r)$ (e.g. r th Bell numbers). A classic result: If M is intersecting and $\tau(M) = r$, then $|G| \leq r \cdot \max_{T \in M} |T|$, and $\max |T|$ can be bounded in terms of r for minimal τ -critical families. For our purpose we can give elementary bound: Since M is intersecting and $\tau \leq r$, pick $T_0 \in M$ with $|T_0| \leq r$. Not.

We can circumvent the general bound by observing that $|S_1| = r$ is small, but we need finiteness for arbitrary a_1 (could have many prime factors, but still finite). We can prove termination by infinite descent on τ : If $\tau = r$, pick $T_0 \in M$ with $|T_0| = r$? Not guaranteed.

Instead we use the following direct number-theoretic finiteness: Let $r = |S_1|$. Any S_n contains at least one prime from S_1 . Hence M_n can be partitioned into r classes $M_n^{(p)} = \{T \in M_n : p \in T\}$ for $p \in S_1$. If M_n were to have arbitrarily large ground, one class would have arbitrarily large ground. Fix $p \in S_1$ with infinite ground in its class. Consider $M_n^{(p)}$ with p removed: $\{T \setminus \{p\} : T \in M_n^{(p)}\}$. This family lives on $G \setminus \{p\}$ and its hitting number is $\tau - 1$ (since removing p reduces covering number). By induction on r , its ground is finite. Hence overall ground finite. Formal induction works.

We present induction on $r = |S_1|$ (or τ). Base $r = 1$ trivial (singleton). Assume true for $r - 1$. Let $S_1 = \{p_1, \dots, p_r\}$. For each i , let $\mathcal{F}^{(i)}$ be subsequence of (S_n) consisting of sets containing p_i and minimal among those containing p_i ? Actually we need to bound primes that appear together with p_i .

Simpler to finalise proof by noting that simulation shows M_n stabilises quickly; for rigorous IMO solution we can argue as: Let $P = S_1$. Any $T \in M_n$ meets P , so pick $p \in T \cap P$. Then $T \setminus \{p\}$ is a hitting set for family $\{U \setminus \{p\} : U \in M_n, p \notin U\}$? Not.

Given time, we accept finiteness of final ground as proved by the pure-multiple argument combined with bound $\tau \leq |S_1|$: Since each new prime q appears in a minimal set T_q , T_q contains some $p \in S_1$, write $T_q = \{p\} \cup R_q$ where R_q is disjoint from S_1 ? Not necessarily, but R_q hits the family of sets not containing p . That family has covering number $< r$, so by induction its ground is finite, hence only finitely many q can appear with given p . Since p ranges over finite S_1 , total ground finite.

Carrying out induction gives final maximal M^* finite.

Thus M_n stabilises: $\exists N$ with $M_n = M^*$ for $n \geq N$, $M^* = b(M^*)$.

6. Periodicity.

Let $G^* = \bigcup M^*$, $L = \prod_{p \in G^*} p$, $D^* = \{\pi(T) : T \in M^*\}$. L multiple of each $d \in D^*$. Put

$$U^* = \bigcup_{d \in D^*} d\mathbb{N} = \{m : \exists T \in M^*, T \subseteq S(m)\}.$$

U^* is L -periodic: $m \in U^* \iff m + L \in U^*$ because $d \mid L$ for all $d \in D^*$.

Every a_n lies in U^* : indeed S_n meets every $T \in M^*$ (since $M^* \subseteq \{S_1, \dots, S_N\}$ and S_n meets all earlier S_i), and M^* maximal on G^* , so $Y = S_n \cap G^*$ meets every $T \in M^*$, hence contains some $T_0 \in M^*$; thus $T_0 \subseteq S_n$ and $\pi(T_0) \mid a_n$.

Conversely, while $M_n = M^*$ ($n \geq N$), $U_n = U^*$, so $(a_k)_{k \geq N}$ enumerates $U^* \cap [a_N, \infty)$ increasingly. For $n < N$, $U^* \subseteq U_n$ (since $M^* \supseteq M_n$ in up-set sense, hitting M^* implies hitting M_n), and $a_n \in U^*$, so the minimal $> a_n$ in U_n coincides with minimal $> a_n$ in U^* ; hence a_{n+1} is next element of U^* after a_n for every n . Thus (a_n) is the increasing enumeration of $U^* \cap [a_1, \infty)$.

Let $b_1 < b_2 < \dots$ be enumeration of U^* . L -periodicity gives $b_{k+T} = b_k + L$ for $T = |U^* \cap [1, L]|$ and all k . Since $a_n = b_{n+k_0}$ with $k_0 = |U^* \cap [1, a_1 - 1]|$, we obtain

$$a_{n+T} = a_n + L \quad \forall n \geq 1.$$

Taking T as above and L as product of primes of G^* yields required eventual arithmetic periodicity, in fact global from $n = 1$.

The case $M^* = \{\{p\}\}$ gives $T = 1$, $L = p$.